

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 11: Product Category Analysis

Aim

To analyze product category sales data using a **funnel chart, table, pie chart, and Tableau dashboard** for interactive exploration of sales across different product categories.

Dataset

Product Categories

Category	Sales (\$)
Electronics	50,000
Clothing	35,000
Appliances	40,000

Q1. Funnel Chart — Sales Conversion Process

Question

Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

R Program

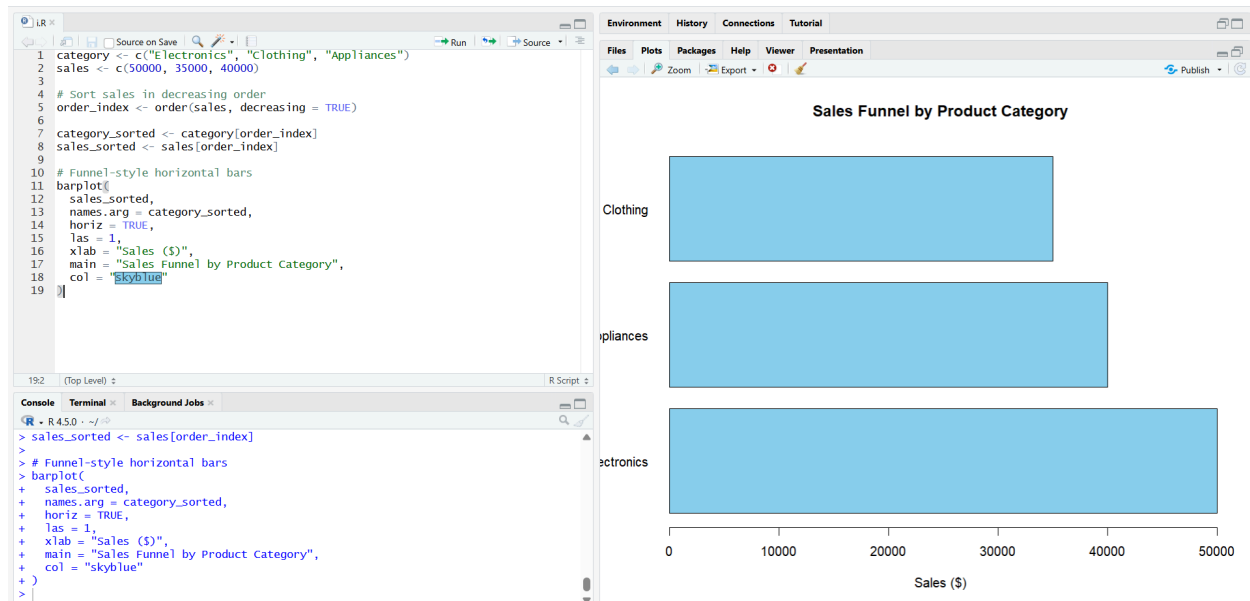
```
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)

# Sort sales in decreasing order
order_index <- order(sales, decreasing = TRUE)

category_sorted <- category[order_index]
sales_sorted <- sales[order_index]

# Funnel-style horizontal bars
barplot(
  sales_sorted,
  names.arg = category_sorted,
  horiz = TRUE,
  las = 1,
  xlab = "Sales ($)",
  main = "Sales Funnel by Product Category",
  col = "skyblue"
)
```

Output



Observation

Electronics has the highest sales, followed by Appliances and Clothing.

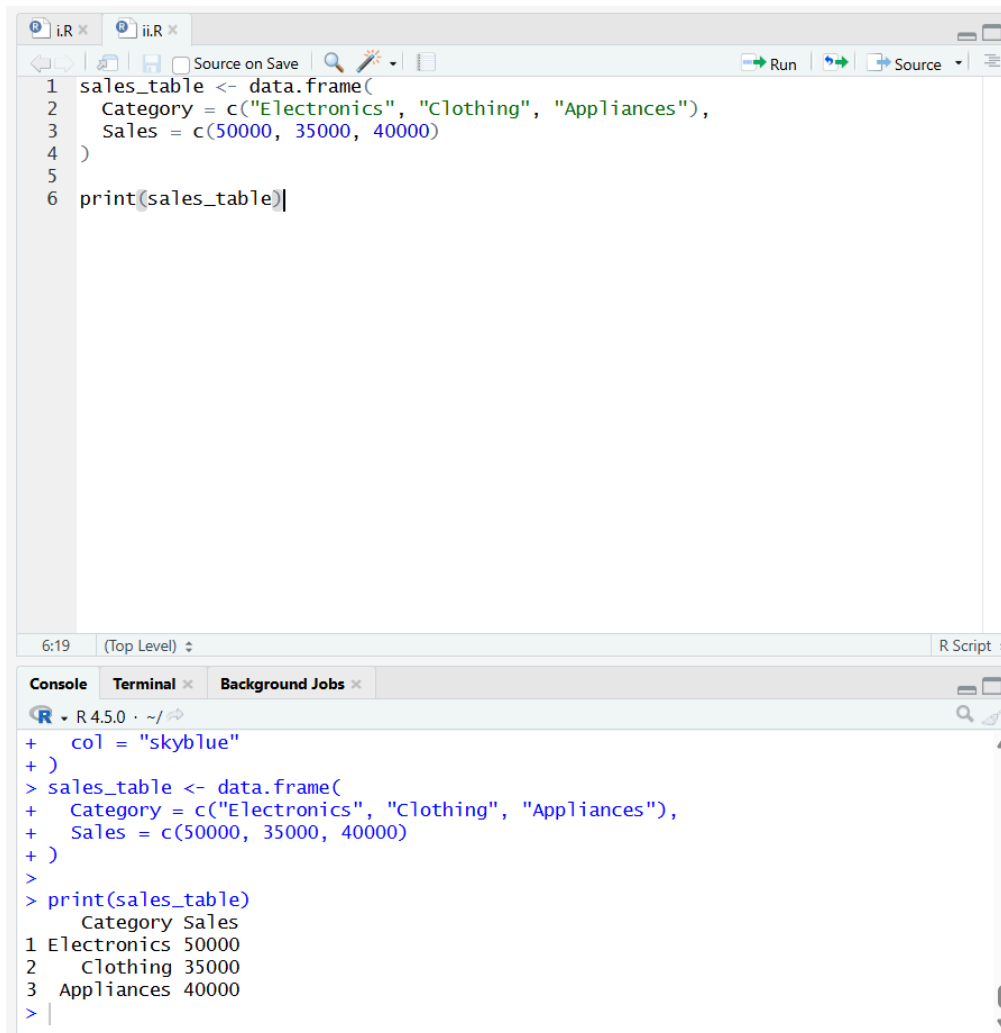
Note: The dataset contains sales by category but does not contain actual conversion stages such as visitors → leads → purchases. Therefore, this chart represents the categories as funnel stages for the purpose of the practical.

Q2. Table — Product Category Sales

R Program

```
sales_table <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)  
  
print(sales_table)
```

Output



The screenshot shows the R Studio environment. The top pane displays the R script with the following code:

```
1 sales_table <- data.frame(  
2   Category = c("Electronics", "Clothing", "Appliances"),  
3   Sales = c(50000, 35000, 40000)  
4 )  
5  
6 print(sales_table)|
```

The bottom pane shows the console output, which includes the execution of the script and the resulting table:

```
+ col = "skyblue"  
+ )  
> sales_table <- data.frame(  
+   Category = c("Electronics", "Clothing", "Appliances"),  
+   Sales = c(50000, 35000, 40000)  
+ )  
>  
> print(sales_table)  
  Category Sales  
1 Electronics 50000  
2  Clothing 35000  
3 Appliances 40000  
> |
```

Observation

The table clearly displays the sales amount for each product category.

Q3. Tableau Dashboard

Question

Develop a Tableau dashboard combining the **pie chart, funnel chart, and table** for interactive exploration of product category data.

Tableau Procedure

Step 1 — Import Data

1. Open **Tableau**.
2. Select **Microsoft Excel** or **Text File**.
3. Import the Product Categories dataset.

Step 2 — Create Pie Chart

- Drag **Category** → Color.
- Drag **Sales** → Angle.
- Drag **Sales** → Label.
- Select **Pie** from the Marks card.
- Title: **Sales Distribution by Product Category**

Step 3 — Create Funnel Chart

Tableau does not provide a standard funnel chart in the basic chart list, so create a funnel-style visualization using sales values.

- Sort categories by Sales in descending order.
- Use Category and Sales to create the funnel-style visualization.
- Title it **Sales Funnel by Product Category**.

Step 4 — Create Sales Table

Add:

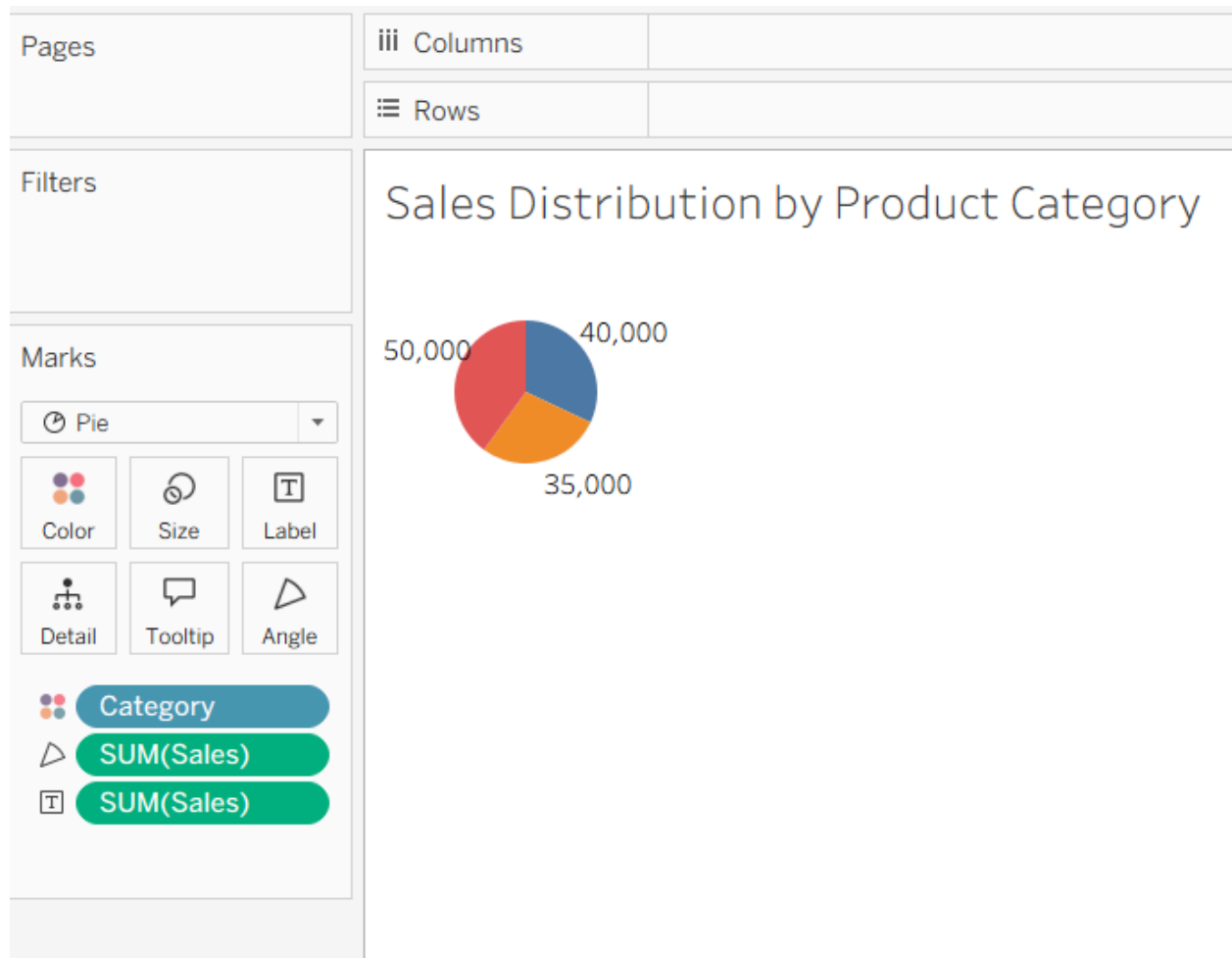
- Category
- Sales

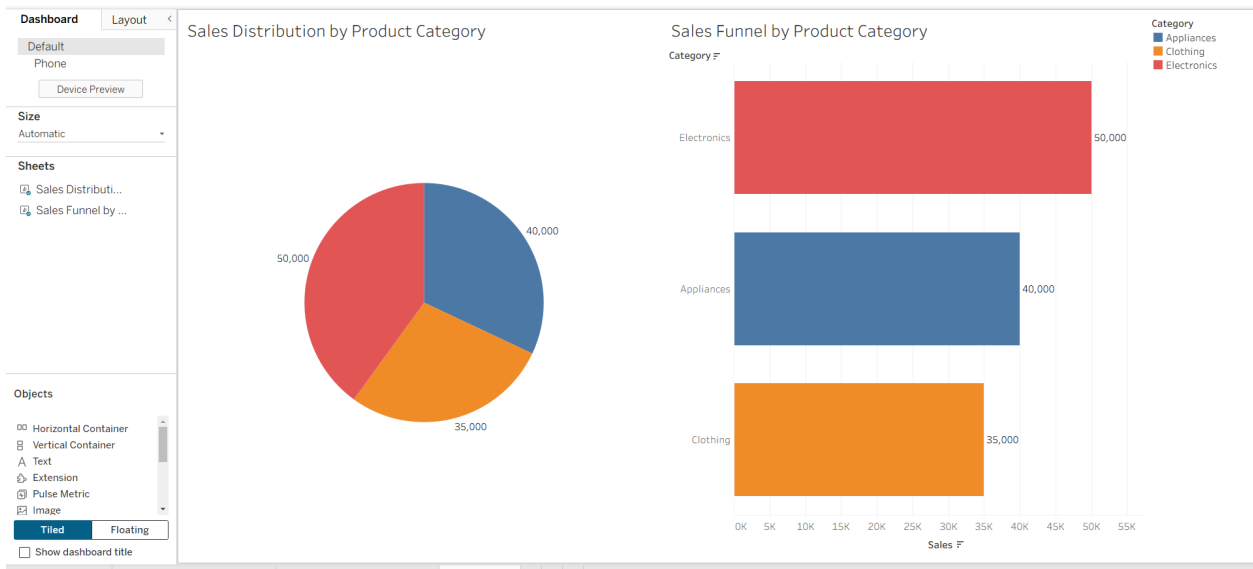
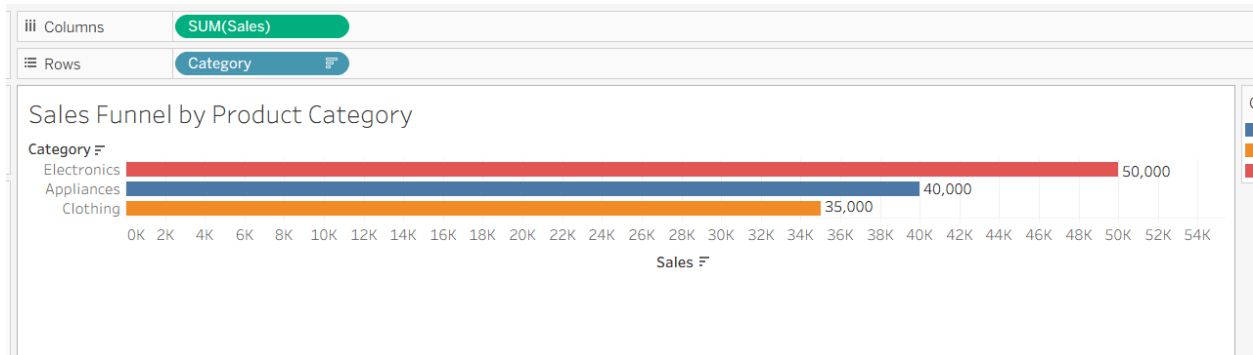
to the worksheet and format the sales values as currency.

Step 5 — Create Dashboard

1. Select **New Dashboard**.
2. Add the **Pie Chart**.
3. Add the **Funnel Chart**.
4. Add the **Sales Table**.
5. Add filters if required.
6. Arrange and format the dashboard.

Expected Dashboard





Q4. Pie Chart — Sales Distribution

Question

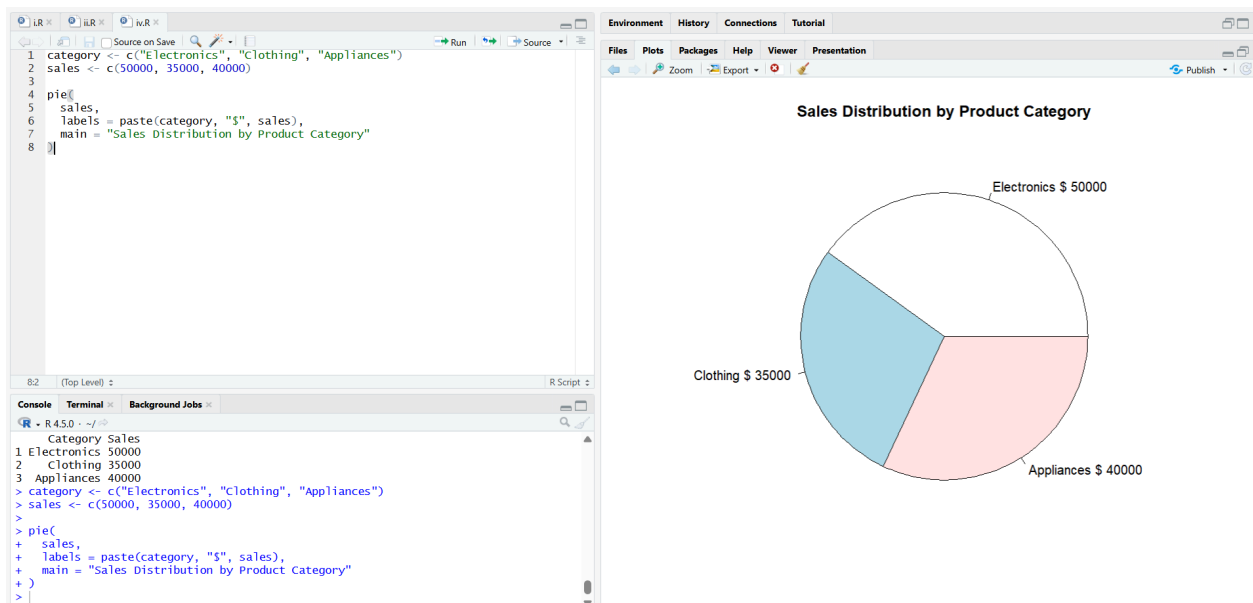
Create a pie chart to represent the distribution of sales across product categories. Include labels.

R Program

```
category <- c("Electronics", "Clothing", "Appliances")  
sales <- c(50000, 35000, 40000)
```

```
pie(  
  sales,  
  labels = paste(category, "$", sales),  
  main = "Sales Distribution by Product Category"  
)
```

Output



Observation

- **Electronics** has the highest sales at \$50,000.
- **Appliances** has \$40,000 in sales.
- **Clothing** has the lowest sales at \$35,000.

Total sales = **\$125,000**.

Procedure

1. Open **RStudio** and create a new R Script.
2. Enter the given product category and sales data.
3. Sort the sales values and use `barplot()` to create a **funnel-style chart**.
4. Use `data.frame()` to create a table displaying the category and sales values.
5. Use the `pie()` function to create a **pie chart showing sales distribution**.
6. Open **Tableau** and import the product category dataset.
7. Create the pie chart using Category and Sales.
8. Create a funnel-style chart using the category sales values.
9. Create a table displaying each category and its corresponding sales.
10. Create a new Tableau dashboard and combine the **pie chart, funnel chart, and table**.
11. Add suitable titles, labels, formatting, and filters for interactive exploration.

Result

The product category sales data was successfully analyzed using **R and Tableau**. A funnel-style chart was created to compare category sales, a table was created to display the sales data, and a pie chart was created to visualize the sales distribution. These visualizations were successfully combined into an **interactive Tableau dashboard** for exploring product category sales.